

Accelerator | Annual Technical Report 2025

Digital Transformation



Drone sprayers used over rice fields.
Credits: IRRI.

Author: Digital Transformation Accelerator

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Collaborating Centers



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CGIAR Technical Reporting 2025

This 2025 Annual Technical Report forms part of CGIAR's first reporting cycle under the [System Council-approved 2025–2030 Portfolio](#). It is grounded in the [CGIAR Research and Innovation Strategy Program/Accelerator proposals](#) and Plans of Results and Budget (PORBs) that establish each Program's and Accelerator's scope and objectives. It aligns with the [Performance and Results Management Framework \(PRMF 2022–2030\)](#), and the [Technical Reporting Arrangement \(TRA 2025–2030\)](#).

An [Accelerator inception report](#) was developed in 2025 to document initial Accelerator design, scope, and implementation arrangements. This report provides important context for interpreting early progress and implementation choices reported in this Annual Technical Report.

Pooled ([Window 1/Window 2, or W1/W2](#)) results are presented alongside mapped Window 3 (W3) and bilateral results. All pooled (W1/W2) results undergo [quality assurance](#) according to established standards. W3 and bilateral results are quality-assured at Center level and reviewed by Programs and Accelerators.

All data presented in this report reflect the status as of 30 March 2026 and draw on quality-assured results in the [Results Dashboard](#), unless otherwise stated. While every effort has been made to ensure data accuracy, note that further refinements are taking place and updates will be available via the CGIAR Results Dashboard.

1. Overview for 2025

Accelerator title	Digital Transformation
Accelerator director	Ram Dhulipala
Start–end date	2025-2030
Centers involved	AfricaRice, Alliance of Bioversity International and CIAT (ABC), CIMMYT, CIP, ICARDA, ICRISAT, IFPRI, IITA, ILRI, IRRI, IWMI, WorldFish
Targeted Impact Areas	Climate Adaptation and Mitigation; Poverty Reduction, Livelihoods and Jobs; Nutrition, Health and Food Security; Gender Equality, Youth and Social Inclusion
Geographic focus	Global and regional (Central America, Eastern and Southern Africa, Latin America, Southeast Asia, Western and Central Africa, Western India Ocean and Asia). The Accelerator's main objective is to create digital and AI tools and solutions that have a cross-cutting impact across various Programs, and therefore a large amount of the Accelerator's work has global focus.
Mapped W3/bilateral projects	14 W3/bilateral projects mapped. The detailed list of W3/bilateral projects mapped to Digital Transformation is available on pages 8-9 of the 2025 Plan of Results and Budget .
2025 budget	US\$24.2M (W1/2: US\$6.6M; W3/bilateral: US\$17.6M)

About the Accelerator

The Digital Transformation Accelerator advances CGIAR's ability to address food, land, and water (FLW) challenges by leveraging data, digital innovations, and artificial intelligence (AI) in an inclusive and ethical manner. As conventional research workflows increasingly depend on data interoperability and AI-readiness, the Accelerator "accelerates" CGIAR science by co-designing and embedding inclusive digital innovations into the work of other Programs, Accelerators and partners. It operates through four areas: Data Ecosystem (FAIR and AI-ready data), Action Lab (co-developed innovations), Digital Futures (emerging technologies), and Enabling Environment (policy and governance). As a cross-cutting Accelerator, it strengthens the digital backbone of CGIAR and amplifies System-wide impact.

Executive summary

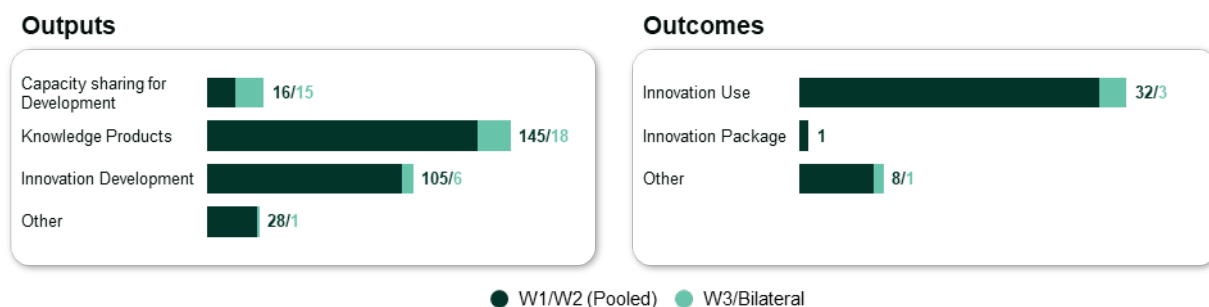


Figure 1. Results reported by the Digital Transformation Accelerator in 2025, by source of funding and result type. Based on data accurate as of 30 March 2026.

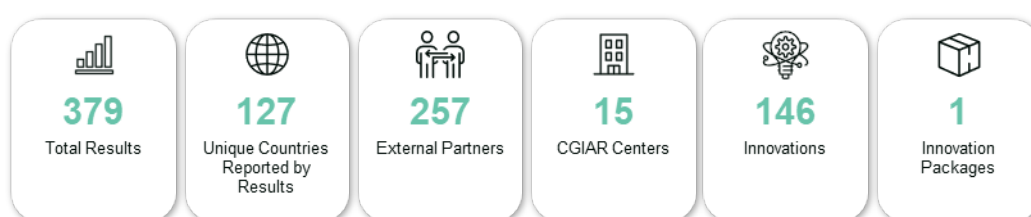


Figure 2. Key performance indicators from 2025 reported results. Based on data accurate as of 30 March 2026.

	Indicators	Met Annual Target	Achievement	Total Results
● AOW01 Data Ecosystem	12	11	92%	98
● AOW02 Action Lab	9	9	100%	106
● AOW03 Digital Futures	5	4	80%	58
● AOW04 Enabling Environment	11	9	86%	110
Total	37	33	89%	372

● On track (≥80% of target achieved)
● Moderate (50-79% of target achieved)
● Behind (less than 50% of target achieved)

Ratings are based on annual target achievement of standard indicator categories within the Theory of Change (ToC). AI-generated ratings were reviewed and adjusted as needed by the Accelerator, with any updates accompanied by justifications.

Figure 3. Accelerator's progress against 2025 ToC targets, by Area of Work. Based on data accurate as of 30 March 2026.

Setting the stage: A year of foundations and firsts

The Accelerator launched in 2025 as CGIAR's dedicated engine for embedding AI and digital innovation across food, land, and water systems research. Operating across four interconnected Areas of Work (AoW) — Data Ecosystem, Action Lab, Digital Futures, and Enabling Environment — the Accelerator focused on building the infrastructure, partnerships, and knowledge base required to accelerate responsible digital transformation. The first year demonstrates both the breadth of engagement across CGIAR Centers and early signs of impact in priority areas.

Flagship achievements: AI tools and data systems gain traction

The year's most prominent milestone was the development of AgriLLM, an open-source large language model trained on CGIAR's agricultural knowledge base and more than 7,400 human-generated question-and-answer pairs, created through a collaboration of more than 15 organizations. Alongside AgriLLM, the CGIAR AI Hub was launched in Abu Dhabi in December 2025, establishing the first dedicated node connecting CGIAR's scientific portfolio with the United Arab Emirates' (UAE) AI ecosystem. The hub is developing products spanning advisory services, genetic resource management, water management, and livestock systems, suggesting unambiguous progress in realizing the Accelerator's vision of creating a global network of digital innovation hubs.

On the data infrastructure side, the Carob framework — which standardizes and aggregates legacy agricultural research data — enabled the creation of a harmonized database of 2,296 original datasets containing more than 2 million records, now used across multiple Programs and bilateral projects. Complementing this, a Data Harmonization Workshop brought together members of 12 CGIAR Centers to develop practical approaches for FAIR and AI-ready data publishing. The GARDIAN platform was redesigned in 2025 to improve navigation, simplify search queries, and incorporate Carob-standardized datasets, strengthening open access to agricultural data assets across the System.

Digital advisory services: Reaching farmers at scale

Several digital advisory innovations demonstrated measurable farmer-level outcomes. In Indonesia, a study of 216 rice farmers found that adoption of the Layanan Konsultasi Padi digital advisory service reduced nitrogen use by an average of 18 kilograms per hectare while increasing yields by approximately 600 kilograms per hectare, with 78 percent of users willing to continue and 76 percent willing to recommend the service. Indonesia's Ministry of Agriculture further embedded the tool by training 17 master trainers through a cascade model for national extension delivery. In India, more than 6,000 farmers in Odisha used the Intelligent Agricultural Systems Advisory Tool (iSAT) climate information services platform for climate-informed farm management. The Meghdoot agrometeorological advisory application, downloaded more than 400,000 times since 2021, received significant infrastructure upgrades in 2025, and the India Meteorological Department (IMD) adopted a CGIAR-developed dashboard to monitor app usage patterns at district and state levels, demonstrating institutional uptake of digital tools.

Building toward inclusive and ethical digital transformation

A defining theme of 2025 was the Accelerator's commitment to responsible, human-centered innovation. The Multidimensional Digital Inclusivity Index online assessment platform was developed to enable structured, participatory evaluation of digital innovations, while a chatbot prototype was created to guide users through the assessment process and standardized mobile-based data collection tools were built to support inclusive digital assessment.

The Digital Futures AoW advanced CGIAR's capacity to anticipate and test emerging digital innovations through a foresight workshop in Colombia involving 22 members from six CGIAR Centers, producing scenario frameworks to guide responsible digital investment decisions. A complementary workshop in Sri Lanka brought together CGIAR researchers to identify priority AI Co-Scientist use cases, including Ask-the-Data and AI-assisted scientific reporting, helping shape the next phase of experimentation with AI-enabled research workflow. Drawing on more than 25 active probes, tested across eight CGIAR Centers to explore frontier digital technologies in safe-to-fail environments before further development or scaling, AoW3 published a brief documenting its PROBES approach for testing emerging technologies.

Challenges and adaptive responses

The year also exposed important constraints. Field evaluations of low-cost portable soil fertility sensors in Guatemala and Colombia revealed insufficient accuracy for precision fertilizer management, demonstrating the value of early-stage probes in preventing premature investment. Budget constraints limited activities on some platforms, including the iSAT climate services in India, underscoring the Accelerator's need for sustained and diversified financing. Also, the operating model of funding meant that many research activities were fragmented across Centers. These challenges have been discussed in the Accelerator's Reflect and Adapt sessions, and proper measures will be identified in future years.

Looking forward

By the end of the year, the Accelerator had established the AI Hub in Abu Dhabi, launched AgriLLM as a flagship collaborative product, advanced data standardization infrastructure across 13 Centers, and built a growing body of evidence on responsible digital innovation. With foundational systems in place, the Accelerator is positioned to deepen its contributions to equitable, AI-enabled agricultural transformation in the years ahead.

2. Annual progress

2.1. Progress along the theory of change

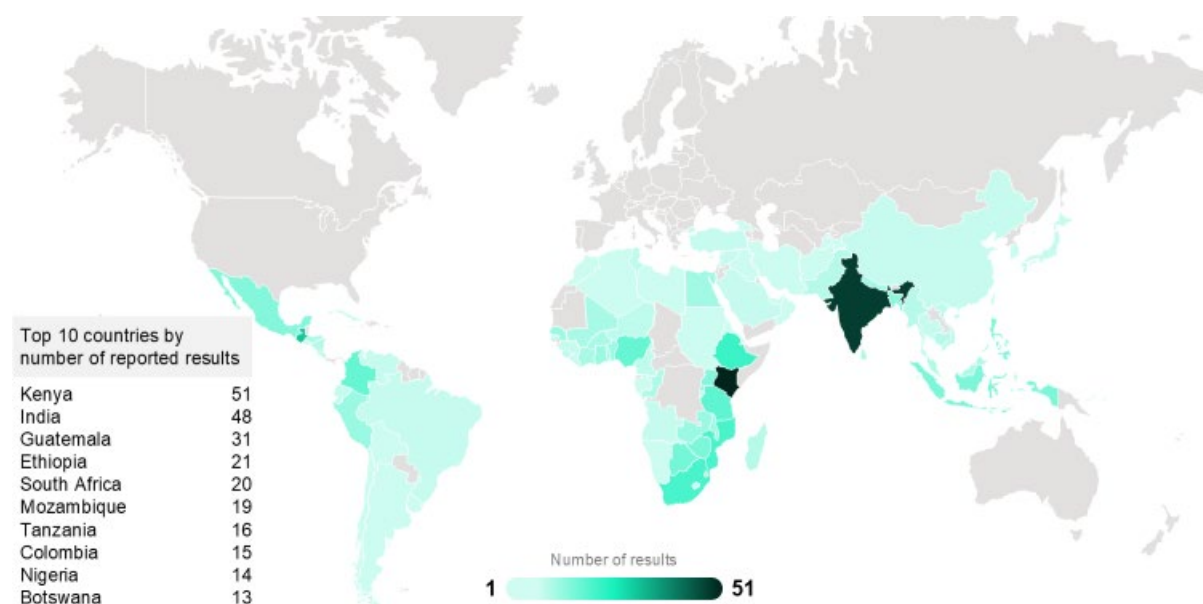


Figure 4. Geographic distribution of 2025 results. Based on data accurate as of 30 March 2026.

Digital Transformation Accelerator: Theory of change, 2025

Building the foundations for responsible digital transformation

The Accelerator made substantial progress in 2025 toward its Theory of Change, which emphasizes that open, standardized data, strengthened capacities, and enabling governance allow scientists and partners to deploy ethical, inclusive, and scalable digital innovations across food, land, and water systems. The focus for 2025 was on establishing foundational infrastructure, co-developing priority tools, and initiating shifts toward data-driven and AI-enabled research.

Strengthening the data ecosystem

Progress was made in advancing open, standardized, and AI-ready data across CGIAR. A [data harmonization workshop](#) produced [shared guidelines for FAIR data publishing](#). The Carob framework enabled the [harmonization of 2,296 datasets with more than 2 million records](#) and was adopted across multiple Programs and bilateral projects. At the same time, [improvements to the GARDIAN platform](#) enhanced data discovery and access, while the File Manager platform supported secure data sharing.

Co-developing AI-powered tools and inclusive innovations

Progress was strong in developing AI tools that enhance research and extend advisory services to farmers. [AgriLLM was developed through a collaboration of more than 15 organizations](#) using over 7,400 human-generated question-and-answer pairs from CGIAR staff. The CGIAR AI Hub [was launched in Abu Dhabi in December 2025](#) to connect CGIAR's scientific portfolio with the UAE's AI ecosystem. In Indonesia, a study of 216 rice farmers showed that [adoption of the Layanan Konsultasi Padi digital advisory service increased yields by an average of 600kg/hectare and net income by 3–5 million Indonesian rupiah per hectare](#). More than 6,000 farmers in Odisha, India, used the [ISAT climate information service](#) to manage climate risks, while the [India Meteorological Department adopted a CGIAR-developed dashboard](#) to monitor advisory app usage nationally.

Anticipating digital futures and building enabling environments

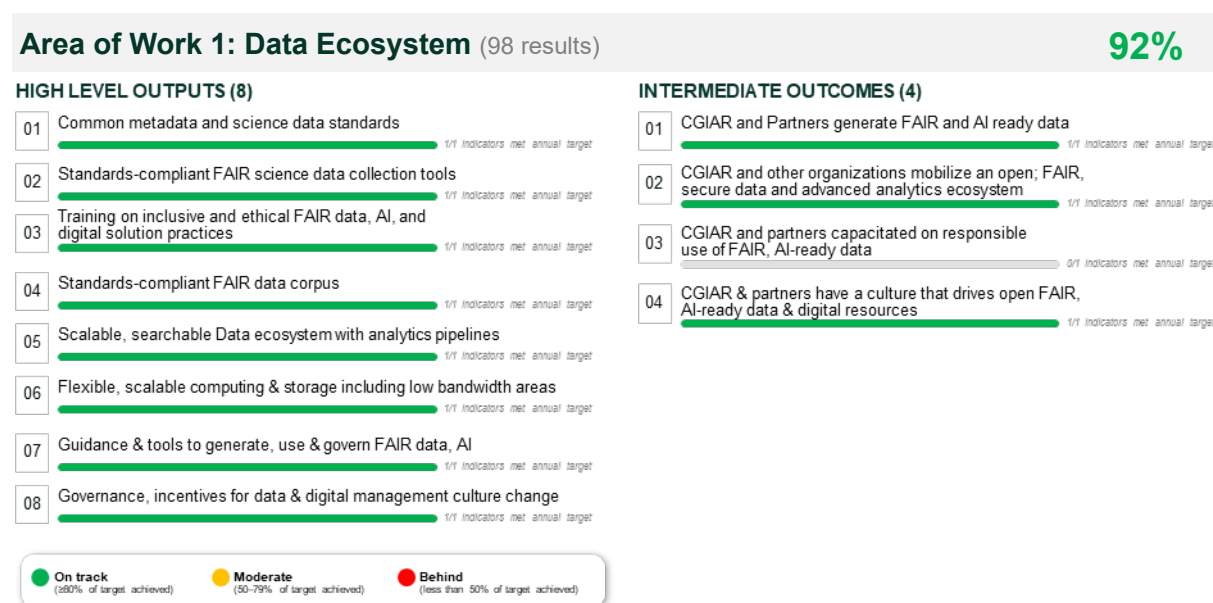
The Digital Futures Area of Work advanced CGIAR's foresight capacity. **Twenty-two members from six CGIAR Centers were trained in participatory foresight methods** to explore how emerging technologies could reshape food, land, and water systems in the Global South. A CGIAR-wide probe survey **catalogued more than 25 active early-stage experiments** across Centers, including probes on voice-based data collection, low-cost soil sensors, and computer vision for market monitoring. On the enabling environment side, the Multidimensional Digital Inclusivity Index **was developed with an online assessment platform** and an AI-assisted conversational tool to help organizations evaluate and improve the inclusivity of their digital innovations. The International Potato Center (CIP) **institutionalized a digital transformation strategy** committing to findable, accessible, interoperable, and reusable (FAIR) data and responsible AI practices by 2030, and the Asia Digital Hub **was established at WorldFish in Penang, Malaysia**, creating the first regional node for digital agricultural innovation in Asia.

Contributing to CGIAR's 2030 Impact Areas

These results contribute meaningfully to **CGIAR's 5 Impact Areas**. Farmer-facing digital advisory tools are improving food security and nutrition outcomes, while inclusive design approaches, including **human-centered design integration across projects in Colombia, Guatemala, and Kenya**, are advancing gender equality and social inclusion. **Digital twins for the Limpopo River Basin and flood early warning in Ethiopia** support climate adaptation, and **four major river basin organizations formally requested Digital Twin support** following the Limpopo success, signaling growing demand for replication.



2.2. Area of Work progress



Ratings are based on annual target achievement of standard indicator categories within the Theory of Change (ToC). AI-generated ratings were reviewed and adjusted as needed by the Accelerator, with any updates accompanied by justifications.

Figure 5: AoW 1 progress against 2025 ToC indicator targets.

Building the foundations for FAIR and AI-ready agricultural research

Standards and tools for interoperable data

In 2025, the Data Ecosystem AoW made substantial progress establishing the infrastructure, standards, and tools needed to make agricultural research data findable, accessible, interoperable, and reusable (FAIR) across CGIAR and its partners. A central achievement was the development and publication of standardized metadata and data guidelines, including the [CGIAR data harmonization guidelines for core variables](#), the [TermInAg controlled vocabulary for agricultural research data](#), and a [Data Harmonization Workshop report](#) that brought together members of 12 CGIAR Centers to build consensus around core data standards for agronomy and socioeconomics domains.

A diverse suite of digital tools was developed or adapted to support standards-compliant data collection and management, including the [Carob framework](#) for standardizing legacy agricultural data, the [Vocal R package](#) for checking standards compliance, the [yuri R package](#) for integrating open research data into reproducible workflows, an [Open Data Kit](#)—compliant form template mapped to TermInAg variables, and a gamified web-based tool for crowdsourcing question-and-answer pairs to support training of the agriculture-focused large language model. Additional tools addressed specific needs, including a satellite remote sensing system for near real-time monitoring of irrigated areas, a mobile app for assessing livestock feed resources, data collection tools for the Multi-Dimensional Inclusivity Index, and the e-Agrology research trials module.

Scaling the FAIR data corpus and analytics platforms

The year saw significant growth in the body of FAIR-compliant datasets. Through the Carob project alone, 57 agronomy datasets, 30 crop variety datasets, 18 soil datasets, and 12 survey datasets were curated under standardized protocols. The FAIRification of 76 legacy datasets by CIP across breeding, agronomy, and plant health domains demonstrated institutional commitment to FAIR-by-design approaches. Further additions included 41 participatory on-farm trial datasets from five countries, a historical agronomic dataset from Mexico spanning 2012–2022 and 598 question-and-answer pairs on livestock agriculture, crowdsourced at a hackathon in Nairobi.

Operational platforms also expanded. The GARDIAN platform was redesigned with improved search and navigation, reaching 31 partners and Program areas. Other deployed platforms included a searchable catalog of rice research datasets, livestock feed composition databases for Sub-Saharan Africa and the Collaborative GARDIAN Labs high-performance computing infrastructure, upgraded to support large language modeling with 24 partner institutions.

Capacity strengthening and institutional adoption

Capacity building reached 77 individuals trained through the Carob training for CGIAR members, 20 researchers trained on responsible AI use, and 18 country office staff trained on inclusive data management practices at the International Rice Research Institute (IRRI). Guidance products included publications on responsible AI use in research, digital transformation of fisheries data systems in Egypt, multiple guidelines on digital data collection, and operationalizing FAIR principles at CGIAR.

Institutional adoption of FAIR practices advanced meaningfully: 10 institutions formally adopted strategies promoting open, FAIR, and AI-ready data practices through the TerminAg vocabulary, and five government institutions across West Africa adopted standardized trial protocols for FAIR data collection. The File Manager platform reached 51 users managing and sharing research datasets securely, while Auburn University integrated CGIAR guidance on ethical AI into graduate courses, signaling uptake well beyond the CGIAR system.

West Africa partners adopting FAIR data practices

A concrete example of culture change in action comes from West Africa, where government institutions in Sierra Leone, Togo, Liberia, Nigeria, and Ghana — trained through the West Africa Hub for Soil Health and Fertility — are now applying standardized digital data collection practices and publishing datasets in repositories, generating data suitable for artificial intelligence applications. This result, delivered in partnership with national agricultural research institutes, demonstrates the Data Ecosystem AoW translating standards and tools into changed institutional practice on the ground.

“ This year marked a shift from discussion to action. We secured engagement from all CGIAR Centers and began the practical work of FAIRifying historical datasets while strengthening the tools used to collect the data of the future. These initial agreements were captured in a first version of the Data Harmonization Guidelines, which we expect to evolve into a more extended version and eventually into more formal FAIRification procedures across Centers.

Andrea Gardezabal, AoW1 Lead

Area of Work 2: Action Lab (96 results)

100%

HIGH LEVEL OUTPUTS (7)

- 01 Research integrates AI analytics and digital technologies to enhance modeling and evidence 1/1 indicators met annual target
- 02 Decision support systems integrate a digital twin approach to enable real-time data use 1/1 indicators met annual target
- 03 Breeding programs use AI-powered phenotyping to accelerate trait discovery and breeding cycles 1/1 indicators met annual target
- 04 Genebanks integrate AI-powered tools to improve germplasm management and access 1/1 indicators met annual target
- 05 Advisory products use AI language models to improve relevance and user engagement 1/1 indicators met annual target
- 06 Capacity sharing programs use AI-powered learning platforms to enhance reach and engagement 1/1 indicators met annual target
- 07 Digital innovations use human-centered design and citizen science to enhance gender responsiveness and social inclusiveness 0/1 indicators met annual target

INTERMEDIATE OUTCOMES (2)

- 01 AI-powered tools enhance research, decision-making, and collaboration for researchers, practitioners, and policymakers 1/1 indicators met annual target
- 02 Smallholder farmers, local innovators, women, and youth groups gain equitable access to user-centered, AI-enabled digital services that support learning, innovation, and inclusive participation 1/1 indicators met annual target

● On track (≥80% of target achieved)
 ● Moderate (50-79% of target achieved)
 ● Behind (less than 50% of target achieved)

Ratings are based on annual target achievement of standard indicator categories within the Theory of Change (ToC). AI-generated ratings were reviewed and adjusted as needed by the Accelerator, with any updates accompanied by justifications.

Figure 6: AoW2 progress against 2025 ToC indicator targets.

In 2025, AoW2 Action Lab delivered a broad portfolio of innovations that integrated AI and digital technologies into agricultural research, advisory services, and capacity sharing activities across CGIAR. Transitioning from the Digital Innovation Initiative. In 2025, Action Lab (AoW2) delivered strategically continued core innovations aligned with the priorities of the new CGIAR Research Portfolio, while laying the foundations for new demand-led innovations in consultation with other Programs and Accelerators for 2026 and beyond.

To promote AI for modeling and evidence generation (High Level Output (HLO) 'Research integrates AI analytics and digital technologies to enhance modeling and evidence generation'), Action Lab strengthened agricultural decision-making through a suite of AI-powered tools. These included the IRRI-led [Global Market Intelligence Platform](#) (GloMIP), which expands access to market intelligence for crop investment decisions; International Food Policy Research (IFPRI)-led Statistics from Space, combining Earth observation and AI for rapid crop monitoring in Mozambique; International Crops Research Institute for the Semi-Arid Tropics' (ICRISAT's) MSOIL platform for digital soil mapping and nutrient recommendations in India; and International Livestock Research Institute's (ILRI's) Machine learning analyses of livestock vulnerability and rangeland degradation in East Africa.

For strengthening decision support systems (HLO 'Decision support systems integrate a digital twin approach to enable real-time data use'), Action Lab advanced digital twin pilots across food, land, and water systems. These included International Water Management Institute (IWMI)'s Limpopo River Basin Digital Twin for water management and drought monitoring, a Somalia-focused Digital Twin for flood preparedness, and ILRI's Kapiti system integrating wearable sensors, GPS, and UAV data for livestock management in Kenya.

For AI-powered phenotyping for breeding (HLO 'Breeding programs use AI-powered phenotyping to accelerate trait discovery and breeding cycles'), the Alliance of Bioversity and CIAT developed a smartphone-based digital phenotyping system, which halved data collection time while improving accuracy, and the [Pheno-i](#) drone-based phenotyping platform was validated across multiple crops. ILRI also developed a Machine learning model for livestock performance prediction using sensor-based data.

For integrating AI into genebanks (HLO 'Genebanks integrate AI-powered tools to improve germplasm management and access'), Action Lab supported systems like CIP's Crops-to-Go, which uses large language models to significantly reduce germplasm search time from weeks to days and AI tools from IRRI and CIP that automate accession screening and monitor in vitro viability.

Significant progress was also made in AI advisory services (HLO 'Advisory products use AI language models to improve relevance and user engagement'), particularly through generative AI applications. Chatbot pilots by the International Maize and Wheat Improvement Center (CIMMYT), ILRI, and IRRI delivered localized advisory services on crops and livestock, reaching farmers and extension agents across Africa and Asia. IFPRI's Generative AI for Agriculture (GAIA) project further contributed by releasing curated datasets downloaded over 44,000 times in 2025.

Digitally enhanced capacity sharing (HLO 'Capacity sharing programs use AI-powered learning platforms to enhance reach and engagement') expanded through both in-person and online training initiatives, including CIMMYT's AgroTutor platform, advisory deployment workshops by ICRISAT and ILRI, and IWMI-led digital twin training programs. In parallel, efforts on human-centered design and digital inclusiveness (HLO 'Digital

innovations use human-centered design and citizen science to enhance gender responsiveness and social inclusiveness') introduced tools such as the Multidimensional Digital Inclusiveness Index (led by IWMI) and strengthened gender and inclusion integration in product design (led by the Alliance of Bioversity and CIAT).

Additionally, Action Lab's Outcomes demonstrated how AI and digital innovations are translating into real-world impact. Examples included the adoption of real-time analytics by the India Meteorological Department (IMD) to improve climate advisory delivery and the operational use of Digital Twins in Southern Africa by the Limpopo Water Course Commission (LIMCOM) to enhance water governance.

Across seven HLOs and two Outcomes, the Action Lab met or exceeded all its 2025 targets.

“ In 2025, transitioning from the Digital Innovation Initiative, Action Lab strategically continued core innovations aligned with the priorities of the new CGIAR Research Portfolio, while laying the foundations for new Portfolio of demand-led innovations in consultation with other Programs and Accelerators for 2026 and beyond.

Jawoo Koo, AoW Lead, Action Lab



Women scientist flying an unmanned aerial vehicle (UEA) for drone-based phenotyping in India. Credit: ICRISAT.

Area of Work 3: Digital Futures (58 results)

80%

HIGH LEVEL OUTPUTS (2)

- 01 Probes of a wide range of AI and/or other novel digital solutions to support system-level scenario analysis of digital frontier technology impacts for decision-makers
1/1 indicators met annual target
- 02 Future case scenarios identified for digital transformation in FWL systems in addressing context-specific challenges and opportunities
1/1 indicators met annual target

INTERMEDIATE OUTCOMES (3)

- 01 Scientists, CGIAR and partners use a broader range of novel digital technologies to make decisions and investments
0/1 indicators met annual target
- 02 Enhanced capacity of CGIAR and partners to design, develop, and implement inclusive digital solutions consider
1/1 indicators met annual target
- 02 Increased awareness and understanding of the potential and risks of frontier digital technologies among key stakeholders
1/1 indicators met annual target



Ratings are based on annual target achievement of standard indicator categories within the Theory of Change (ToC). AI-generated ratings were reviewed and adjusted as needed by the Accelerator, with any updates accompanied by justifications.

Figure 7: AoW 3 progress against 2025 ToC indicator targets.

Digital Futures is the Digital Transformation Accelerator's forward-looking AoW, designed to help CGIAR and its partners anticipate emerging technological shifts, critically assess their implications, and guide strategic investments in responsible digital innovation for food, land, and water systems. In 2025, AoW3 contributed 54 results, spanning innovation development, knowledge products, capacity sharing, and other outcomes. These results reflect collaboration across multiple CGIAR Centers and Programs, translating early exploration into tangible outputs and shared learning.

A flagship activity of AoW3 was the Technology Horizons Exploration, which established a safe-to-fail space to probe emerging digital technologies. In 2025, 25 probes were implemented across 8 CGIAR Centers and 6 Programs, generating structured insight into how frontier technologies may shape research and development in FLW systems. These probes explored technologies such as computer vision, sensors and IoT, speech and natural language processing (NLP) interfaces, and large language models (LLMs). Beyond testing individual technologies, the probe portfolio also helped map which Centers are working with which technologies, how these technologies connect to specific Programs, and what types of challenges emerge repeatedly across experiments.

By documenting early experiments and synthesizing lessons across Centers, the probe portfolio provided System-level visibility into emerging digital innovation trends and helped identify technologies with potential to evolve toward prototypes and pilots.

A second flagship activity of AoW3 focused on Living the Digital Futures of Food Land and Water Systems, a foresight initiative designed to explore how digital transformation may unfold under different future conditions. Through participatory workshops involving CGIAR researchers and partners, the initiative co-designed three scenario sets based on high uncertainty and high impact drivers shaping digital transformation. These scenarios provide a structured way to explore alternative pathways for digital innovation and their potential implications for FLW systems. Importantly, the exercise introduced a human-centered perspective to foresight, exploring scenarios through the lived experiences of different personas such as farmers, community leaders, researchers, and policymakers.

AoW3 also advanced the concept of a "CGIAR AI Co-Scientist", which aims to support researchers through AI-assisted workflows and data-driven scientific processes. In 2025, the vision of the AI Co-Scientist was defined through human-centered design sessions, leading to the identification of two priority use cases: "Ask the Data" and a Scientific Reporting Assistant. These use cases explore how AI systems can assist scientists in interacting with complex datasets and synthesizing research outputs. The work in 2025 marked the transition from concept definition toward the early stages of prototype development.

Taken together, these activities position AoW3 as a strategic exploration space within the Accelerator. By combining technology probes, foresight through future scenarios, and experimentation with AI-enabled research approaches, AoW3 helps CGIAR stay ahead of emerging digital trends. The insights generated through these activities support more informed decision-making about where the system should focus future research, experimentation, and strategic investment in digital innovation.

Digital Futures explores emerging digital frontier technologies with the potential to transform FLW systems. Through foresight, safe-to-fail technology probes, and exploration of AI-enabled research workflows such as the AI Co-Scientist, AoW3 helps CGIAR stay ahead of technological trends, anticipate future challenges and opportunities, and generate early evidence to guide strategic digital investments across the CGIAR Research Portfolio.

“ Digital Futures allows CGIAR to stay ahead of the curve by probing frontier technologies and exploring future scenarios that may reshape FLW systems, informing where CGIAR should strategically invest in digital innovation.

Daniel Jimenez, AoW Lead, Digital Futures.



Mobile phones with SMS; with such mobile phones, text messaging and call centers are options for getting information to farmers. Credit: CGIAR.

Area of Work 4: Enabling Environment (110 results)

86%

HIGH LEVEL OUTPUTS (6)

- 01 Frameworks, - methods, and guidelines to catalyze adoption of responsible and inclusive digital and data innovations across FLW systems 0/2 indicators met annual target
- 02 Robust, secure and flexible foundational digital infrastructure & sustainable operating model across centers and hubs 1/1 indicators met annual target
- 03 Networks of digital innovation nodes & hubs creating an ecosystem for inclusive and FAIR compliant digital and data innovations 0/2 indicators met annual target
- 04 Partnership and sustainable business models for digital transformation in FLW systems 1/1 indicators met annual target
- 05 Capacity building & training packages on human-centric digital product design, development, sustainability, and business models 1/1 indicators met annual target
- 06 AI-based Inclusive digital products/Innovations co-created in partnership with the digital innovation nodes 1/1 indicators met annual target

● On track (≥80% of target achieved)
 ● Moderate (50-79% of target achieved)
 ● Behind (less than 50% of target achieved)

INTERMEDIATE OUTCOMES (3)

- 01 Scientists and innovators access methods, tools and policies to generate data that is FAIR and AI ready 0/1 indicators met annual target
- 02 Improved capacity across scientists and innovators to develop and deploy responsible digital innovations and sustainable business models 1/1 indicators met annual target
- 03 Governments adopt policies that promote use of digital innovations in agri-food systems 0/1 indicators met annual target

Ratings are based on annual target achievement of standard indicator categories within the Theory of Change (ToC). AI-generated ratings were reviewed and adjusted as needed by the Accelerator, with any updates accompanied by justifications.

Figure 8: AoW4 progress against 2025 ToC indicator targets..

The Accelerator's Enabling Environment (AoW4) creates institutional and human capacities through supportive policy frameworks, methodologies, knowledge products and capacity building to enable CGIAR Centers and broader agrifood system actors to digitally transform agrifood systems in a responsible, equitable and ethical way. This AoW works through three flagship outputs, which are a Network of Hubs (NOH), a Center of Excellence (COE) and a Digital Core. The AoW's ToC articulates how these three distinct flagship outputs build CGIAR's capacities for responsible and human-centric digital transformation pathways. In 2025, Enabling Environment made great progress across these three flagship activities.

As part of the NOH (HLO 'Networks of digital innovation nodes & hubs creating an ecosystem for inclusive and FAIR compliant digital and data innovations'), the Accelerator launched three digital innovation hubs which are meant to anchor CGIAR's partnerships with tech companies and digital innovators including AgTechs. The three hubs launched in 2025 are the AI Hub (United Arab Emirates (UAE)), the World Fish Hub (Penang, Malaysia) and the global biodiversity hub (Lima, Peru). Amongst the three, the AI Hub in UAE which is being set up with support from the UAE International Affairs Office, facilitates CGIAR's use of the UEA AI Ecosystem and the co-development of digital AI products (HLO 'AI-based Inclusive digital products/Innovations co-created in partnership with the digital innovation nodes') with other Programs in the Portfolio. A Minimum Viable Product (MVP) version of the first product (AgriLLM) was launched in late 2025 while three other AI products are at various stages of development: Genebanks AI (Genebanks Accelerator), Policy AI (Policy Innovations Program) and Livestock AI (Sustainable Animal and Aquatic Foods Program).

Another flagship output of this AoW delivered in 2025 is CGIAR's Digital Transformation Center of Excellence (COE). The COE identified the following practice areas: Human Centered Design (HCD), Responsible AI, Gender and Digital, Digital Economies, and a Consortium for Spatial Imagery. For each of these practice areas, communities of practice (COPs) have been created (HLO 'Frameworks, methods, and guidelines to catalyze adoption of responsible and inclusive digital and data innovations across FLW systems'), and practice leads/convenors have also been assigned. The COPs made substantial progress in 2025 and produced several important knowledge products (HLO 'Partnership and sustainable business models for digital transformation in FLW systems') as well as reusable methodological frameworks (HLO 'Frameworks, methods, and guidelines to catalyze adoption of responsible and inclusive digital and data innovations across FLW systems'). The practice areas also carried out significant capacity building work (HLO 'Capacity building & training packages on human-centric digital product design, development, sustainability, and business models'). Furthermore, these COPs are increasingly leveraged in the context of landscape assessments as well as for the preparation of forward-looking scenarios that are critical to informing CGIAR's AI and digital transformation strategy.

Digital Core, which is the third flagship output of this AoW, strives to embed foundational hard and soft digital capacities within Centers and Programs. In 2025, Digital Core worked closely with two Centers (CIP and IRRI) to develop digital transformation strategies (HLO 'Robust, secure and flexible foundational digital infrastructure & sustainable operating model across centers and hubs') for their respective Centers.

“ The Enabling Environment AoW strives to build the digital innovation muscle of CGIAR and broader agrifood systems.

Ram Dhulipala, AoW Lead, Enabling Environment

3. Portfolio linkages and partnerships

Embedding digital innovation across the CGIAR Portfolio

The Accelerator is designed not to conduct standalone field research but to accelerate the pace and quality of science across the entire CGIAR Portfolio by co-developing, embedding, and scaling inclusive digital innovations. In 2025, this mandate translated into a rich body of results that cut across Programs, research Centers, and external partners — demonstrating the Accelerator's growing role as connective tissue within CGIAR's 2025–2030 Portfolio.

The Accelerator's Portfolio linkages were most visible in its work on data infrastructure. A hybrid, four-day [data harmonization workshop](#) brought together members of 12 CGIAR Centers to [develop shared standards for findable, accessible, interoperable, and reusable \(FAIR\) and artificial intelligence \(AI\)-ready data publishing](#), beginning with agronomy and socioeconomics and gender domains. The resulting [CGIAR data harmonization guidelines](#) for core variables and the first session of the [Data Collaborative](#) mark concrete steps toward the Accelerator's 2030 outcome of institutionalizing FAIR data principles across CGIAR. These efforts were reinforced by the [Carob data standardization framework](#), which in 2025 supported harmonization of 2,296 original datasets containing more than 2 million records, [with adoption extending to national agricultural research institutions in Sierra Leone, Togo, Liberia, Nigeria, and Ghana](#). A dedicated [Carob training](#), organized with the Sustainable Farming Program, further embedded these practices across CGIAR Centers.

AgriLLM: A flagship example of Portfolio co-creation

The development of [AgriLLM](#) — an open-source, agriculture-focused large language model — stands as the most vivid illustration of the Accelerator's co-creation model in action. The initiative was led by IFPRI, with contributions from all CGIAR Centers, drawing on more than 7,400 human-generated question-and-answer pairs produced by CGIAR staff. External partners including Digital Green, the Food and Agriculture Organization of the United Nations, the International Fund for Agricultural Development, the World Bank, the Kenya Agricultural and Livestock Research Organization (KALRO), and universities in Chicago, Florida, and Abu Dhabi all contributed to the model's development. [A gamified tool developed at ILRI in Nairobi](#) was used to crowdsource livestock-relevant question-and-answer pairs from ILRI staff and campus partners, [generating 312 validated pairs for model training](#). AfricaRice contributed a further [531 expert-developed rice question-and-answer pairs](#), and WorldFish developed a dedicated aquatic foods dataset. This CGIAR-wide co-production, described as a flagship product of the Accelerator's Action Lab, directly advances the 2030 outcome of functional partnerships for digital innovation.

The [CGIAR AI Hub](#), [launched in Abu Dhabi in December 2025](#) in partnership with the UAE's AI ecosystem, operationalizes this ambition at institutional scale. Working with CGIAR Centers and Programs, the hub develops AI-enabled products across agricultural advisory services, genetic resource management, water management, and livestock systems, combining CGIAR's scientific depth with external AI infrastructure and expertise.

Strengthening links with Programs and national partners

Linkages with the Breeding for Tomorrow Program were advanced through the [Crops-to-Go](#) germplasm recommendation system, developed at CIP in partnership with Amazon Web Services' generative AI innovation center, and through the [Artemis AI-powered smartphone phenotyping platform](#) developed with Tanzania's Agricultural Research Institute (TARI) and Colorado State University. These tools directly serve breeding programs' needs for faster, data-driven germplasm discovery and field phenotyping.

Links with the Sustainable Farming Program were evident in the [Carob framework's](#) implementation in use cases in Kenya, Zambia, and Ethiopia, and in the Fairgrounds federated data platform, which was [adopted by the Alliance of Bioversity International and CIAT's Tropical Forages Program](#) to streamline machine learning workflows across South America and Southern Africa.

The Sustainable Animal and Aquatic Foods Program benefited from digital twin work on livestock resilience at ILRI's Kapiti Ranch, developed jointly with Scotland's Rural College and technology partner Bodit, which [used wearable sensors and Machine learning to quantify heat stress impacts on dairy cattle](#). The Multifunctional Landscapes Program was served through [digital twin applications for transboundary water management in the Limpopo River Basin](#) and through the operationalization of the [Limpopo Digital Twin for regional water security planning](#).

National research and government partners as co-implementers

National agricultural research and government institutions played active roles in co-implementing and scaling the Accelerator's results. KALRO partnered with IRRI to build national capacity in satellite-based [rice monitoring in Kenya](#), with training covering the full workflow from data preprocessing to yield estimation. In India, the India Meteorological Department adopted a [CGIAR-developed dashboard to monitor the Meghdoot agrometeorological](#)

advisory application across the country, while a new AI-powered agro-advisory project was launched with the Indian Council of Agricultural Research (ICAR)'s Central Research Institute for Dryland Agriculture under India's Mission Mausam Program. Indonesia's Ministry of Agriculture embedded the Layanan Konsultasi Padi digital advisory service into its national extension system through a cascade training model, with 17 master trainers prepared to disseminate recommendations across provinces.

Building a shared enabling environment

The Accelerator's enabling environment work in 2025 generated shared infrastructure and governance assets that benefit the wider Portfolio. A CGIAR AI Readiness and Needs Assessment, involving all 15 Centers, mapped opportunities and risks for responsible generative AI use in research. The Multidimensional Digital Inclusivity Index (MDII) was piloted in a capacity-building workshop in Botswana with representatives of the Limpopo Watercourse Commission's four member states, using the index to evaluate the Water Copilot tool. A participatory foresight workshop in Colombia, convening 22 members from six CGIAR Centers, produced scenario frameworks to guide responsible digital investment decisions across the Research Portfolio, with early reflections from ILRI, CIP, and IFPRI documenting institutional uptake of the workshop's findings.

Together, these results show the Accelerator functioning as its ToC intends: not as a standalone research entity but as a System-wide Accelerator, co-creating the data foundations, AI tools, governance frameworks, and institutional capacities that enable other Programs and national partners to deliver faster, more equitable, and more evidence-based agricultural impact by 2030.

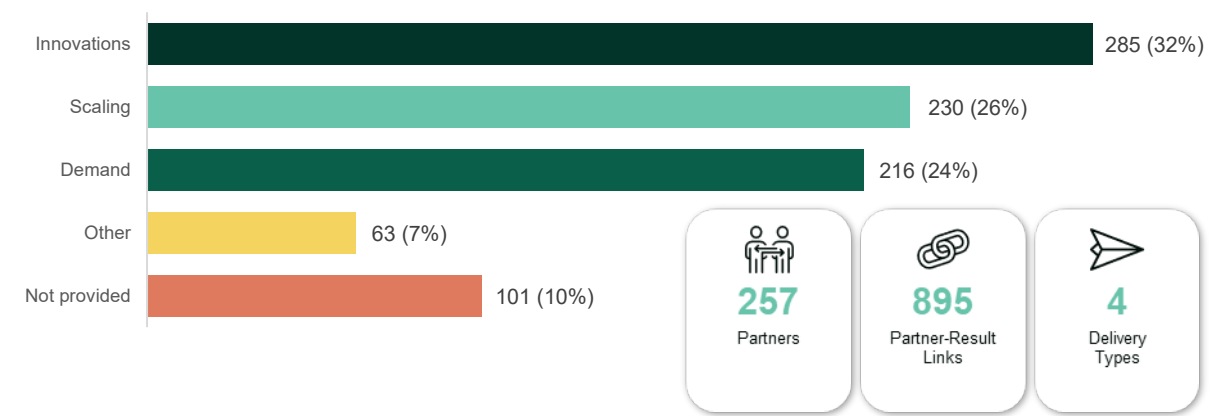


Figure 9: Partner network comprising 257 partners contributing to results across three main delivery types. Innovation-focused collaboration was most prevalent (32%), followed by scaling (26%) and demand (24%).

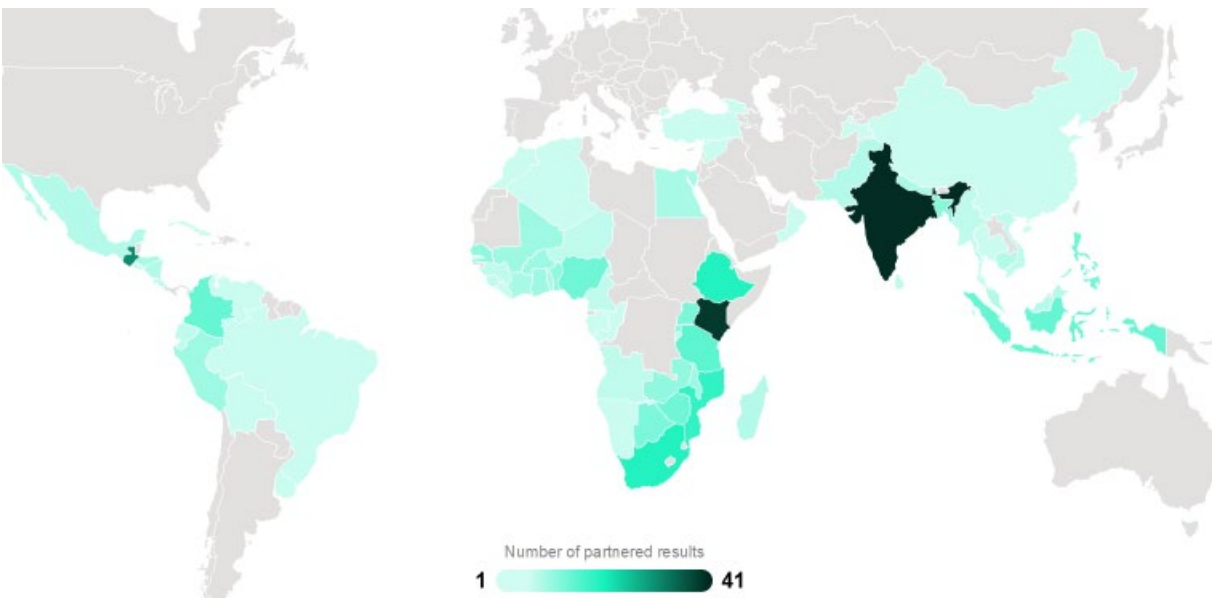


Figure 10: Geographic distribution of results achieved with Digital Transformation's external partners. Based on data accurate as of 30 March 2026.

4. Adaptive management

As a part of adaptive management for 2025, the Accelerator conducted two structured pause and reflect sessions which enabled constructive evaluation of results and evidence in 2025, drawing on performance data (PRMS, Key Performance Indicators (KPIs), tracking), budget and financial analysis, and feedback from ISDC and the Evaluability Assessment (EA). Together, these inputs provided a robust evidence base to assess performance, identify systemic constraints, and define targeted adjustments to strengthen delivery in 2026.

A detailed description of how the Accelerator used the adaptive management process is presented below:

Evidence and learning

Evidence from implementation and performance tracking in 2025 indicates that clearer structuring of work contributed to stronger delivery. AoWs with more focused HLOs were better able to concentrate efforts and align outputs, while overall KPI achievement — particularly in capacity development — demonstrates strong delivery capability across Centers.

At the same time, the evidence highlights important structural gaps. A high volume of unplanned results indicates that HLOs and KPIs did not fully capture the breadth of activities delivered. This reflects misalignment between the ToC and actual implementation, as well as limitations in indicator design. In several cases, planning emphasized specific result types (e.g., innovations), while other outputs (e.g., knowledge products and datasets) emerged during implementation but were not adequately reflected in the results framework.

Additional challenges include inconsistencies in how results are defined and reported, particularly in distinguishing between probes and innovations, overrepresentation of participation-based metrics, and misalignment across AoW. Deliverables were also not consistently or clearly specified in planning modules, limiting visibility and coordination across Centers.

The Evidence and learning (EA) further reinforced these findings and provided clear recommendations to strengthen the Program's analytical depth, coherence, and usability. The EA emphasized the need to move beyond high-level IF-THEN logic and develop a more detailed, step-by-step ToC that clearly articulates assumptions and causal pathways. It also highlighted the importance of strengthening logical frameworks at both Program and AoW levels so they can function as analytical tools rather than descriptive diagrams. Additional recommendations included developing clear evaluation questions to guide indicator selection, distinguishing monitoring from evaluation, integrating key inputs such as capacity development, and ensuring stronger alignment between MELIA, ToC, and budgets.

In response to EA findings, the Accelerator has taken several steps. The MELIA Plan and Inception Report were expanded to better articulate intervention logic and results pathways. Revised logical frameworks were developed and submitted through the ToC board, incorporating improved structure and, where applicable, research questions and assumptions. Supporting materials were shared with the EA team to enhance transparency. Progress has also been made in aligning planning and reporting through the introduction of Package of Deliverables (PoD)-based indicators for 2026.

Building on both implementation evidence and EA recommendations, the Accelerator will further revise its ToC and HLO structure in 2026 as part of the adapt process to better reflect delivery patterns and reduce reliance on unplanned reporting. Similarly, KPI frameworks will be rationalized to ensure realistic, outcome-oriented targets aligned with resources. Clearer definitions and quality thresholds will be introduced across result types, and activities will be increasingly organized around a smaller number of core, shared deliverables.

Looking ahead, these adjustments will strengthen coherence between planning and delivery, improve the credibility and usability of results, and position the Accelerator to better support evidence-based decision-making across CGIAR.

W1/W2 funding and financial planning

The 2025 funding model provided flexibility to Centers, which supported adaptive implementation and responsiveness to emerging opportunities. This flexibility remains a key strength of the Accelerator's operating model.

However, reflection and financial analysis identified several constraints. Budget utilization was uneven, partly due to the compressed timelines of the inception year. More importantly, funding was distributed across many small activities, limiting the ability to concentrate resources on high-impact outputs. The current allocation model also provided limited incentives for cross-Center collaboration, resulting in predominantly decentralized delivery. In addition, weak alignment between budgets, KPIs, and deliverables reduced the ability to strategically direct resources toward priority outcomes.

To address these challenges, the Accelerator will strengthen the linkage between financial planning, KPIs, and ToC priorities as part of 2026 adapt cycle. Resource allocation will be more explicitly aligned with defined deliverables and expected outputs, and greater emphasis will be placed on enabling collaborative delivery models, including multi-Center initiatives and shared deliverables. Going forward, these changes will support a shift from fragmented spending toward more strategic investment, improving both efficiency and impact of resource use.

W3/Bilateral data and alignment

W3 and bilateral projects play an important role in contributing to Accelerator outcomes and extending impact beyond pooled funding. They offer opportunities to complement core investments and strengthen the overall result narrative. However, the pause and reflect sessions highlighted persistent challenges in alignment and reporting.

To address these issues, the Accelerator will focus more on integrating bilaterals into a coherent results narrative, ensuring that pooled and bilateral contributions are aligned and mutually reinforcing.

Risk management

In 2025, key risks related to fragmentation of activities, data governance (integrity, ownership, security), and data culture remained relevant. Fragmentation materialized as a significant risk, affecting focus and limiting delivery of high-impact outputs. Data-related risks were actively mitigated through collaborative efforts across CGIAR in establishing FAIR data systems, governance frameworks, and institutional capacities, improving data management and stewardship across centers. However, challenges in data ownership and data culture continue to constrain full uptake. Risks related to cross-program partnerships and stakeholder coordination were largely minimized through close alignment and collaboration with other science programs and accelerators.

Read Digital Transformation's [Risk Register](#), which includes the top five submitted risks for 2025 and related actions and controls to manage those risks.

5. Key result story

CGIAR AI Hub: Turning agricultural science into AI products

CGIAR's AI Hub transforms decades of cutting-edge agricultural research into practical AI products, placing them directly in the hands of smallholder farmers, policymakers, and researchers.



From left to right: Jawoo Koo, interim Deputy Director, Digital Transformation Accelerator, Bill Gates, and Mariam bint Mohammed Almhiri, Head of the International Affairs Office at the UAE Presidential Court, at the launch of the CGIAR AI Hub in Abu Dhabi in November 2025. Credit: CGIAR.

Hosted in Abu Dhabi and powered by UAE's leading AI ecosystem, the Hub is bridging the gap between research and real-world impact at unprecedented speed. Its purpose is to transform how agricultural research is used by converting decades of science into practical digital products. By working closely with CGIAR Centers and Programs, AI experts, and scaling partners, the Hub enables faster, data-driven decisions for agriculture, climate, and food systems, and strengthens CGIAR's ability to deliver real-world impact.

While artificial intelligence (AI) is widely seen as a transformative solution for agriculture, most organizations struggle to translate this promise into real impact. The 500 million smallholder farmers who feed much of the developing world need timely, context-specific guidance, which can be enabled by AI. Yet, most agricultural knowledge remains locked in academic publications, inaccessible when and where decisions are made. Despite significant global investment in AI, **only 5 percent of organizations report measurable returns** from AI pilots. The challenge is not the technology; it is the translation gap between AI's potential and the people who need it most.

CGIAR is uniquely positioned to close this gap. Over 50 years, its 15 global research Centers have generated authoritative agricultural science, built vast geospatial datasets, and forged partnerships with governments across the Global South. A **2025 survey** conducted by the **Accelerator** identified more than 400 AI-ready use cases across CGIAR Centers. What is needed now is a coordinated effort to turn this comparative advantage into AI-powered products at scale.

Launched in 2025 by CGIAR Digital Transformation Accelerator, the **CGIAR AI Hub** does just that and acts as a catalyst for innovation. Through co-design workshops with CGIAR scientists from all Centers and end-user representatives, the Hub transforms ideas into validated prototypes in shorter cycles. This product-centric approach replaces isolated digital pilot projects with a coordinated, system-wide process, ensuring that innovations move more quickly from experimentation to tangible impact.

The Hub functions as the convergence point of Abu Dhabi's AI Ecosystem for Global Agricultural Development, a unified partnership connecting CGIAR's scientific heritage with the UAE's world-leading AI infrastructure. This ecosystem, established through the UAE–**Gates Foundation** partnership, gives the Hub the capacity to scale rapidly. Through key partnerships, promising concepts are rapidly developed into prototypes. **MBZUAI** (Mohamed Bin Zayed University of Artificial Intelligence) and its newly launched Institute (IA/AI) provide advanced AI research capabilities. **AI71** delivers engineering expertise to develop and deploy the Hub's products.

Each of the Hub's products converts CGIAR science into a practical tool for a defined user group. Five products are currently in active development across different stages.

AgriLLM, a specialized large language model (LLM) for agriculture, **launched in December 2025**, is trained on over 200,000 curated question-and-answer pairs from CGIAR and more than 15 global partners. This high-quality data corpus will allow users, primarily farmers, to receive reliable, context-specific answers to agricultural queries. Supported by partners through rigorous testing and dissemination, the product aims to support over 100 local languages and to provide advice that is twenty percent more accurate than general LLMs.

Genebank AI, an AI-powered platform linking genetic, trait and phenotypic data across CGIAR's eleven Genebanks and its 700,000 accessions, streamlines the search for climate-resilient germplasm. The platform will enable breeders to identify promising lines faster, help researchers navigate vast genomic datasets, and allows Genebank teams to align structured data with real-world user needs. Ultimately, this product will facilitate farmers' access to the resilient varieties required for stable harvests and long-term food security.

Hydrology AI is another AI-driven product that combines near-real-time global datasets with hydrological modelling of river basins, eliminating the need for manual configuration. This product enables the deployment of accurate hydrological models that reflect real-world conditions at scale.

Products like those reduce the time and effort required to analyze agricultural data, strengthen research processes, and enable faster responses to real-world challenges. While researchers are primary users, the benefits extend further to policymakers, development organizations, extension agents, and ultimately farmers, who rely on better information to improve productivity, resilience, and livelihoods.

“ The AI Agriculture Ecosystem is rooted in science and powered by global collaboration. By combining the AI expertise and insights from global partners, we can develop innovations that strengthen decision-making, guide policies and investments, and accelerate the adoption of digital tools, supporting vulnerable communities in the Global South and farmers in the Global North facing similar challenges.

Ismahane Elouafi, CGIAR's Executive Managing Director.

Contributing Centers: AfricaRice, Alliance of Bioversity International and CIAT, CIMMYT, CIP, ICARDA, ICRISAT, IFPRI, IITA, ILRI, IRRI, IWMI, WorldFish

Contributing external partners: Abu Dhabi's AI Ecosystem for Global Agricultural Development (UAE), Gates Foundation, ai71, AIMforScale, Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI).

